

6. Multimodal Visual Understanding + SLAM Navigation

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Introduction

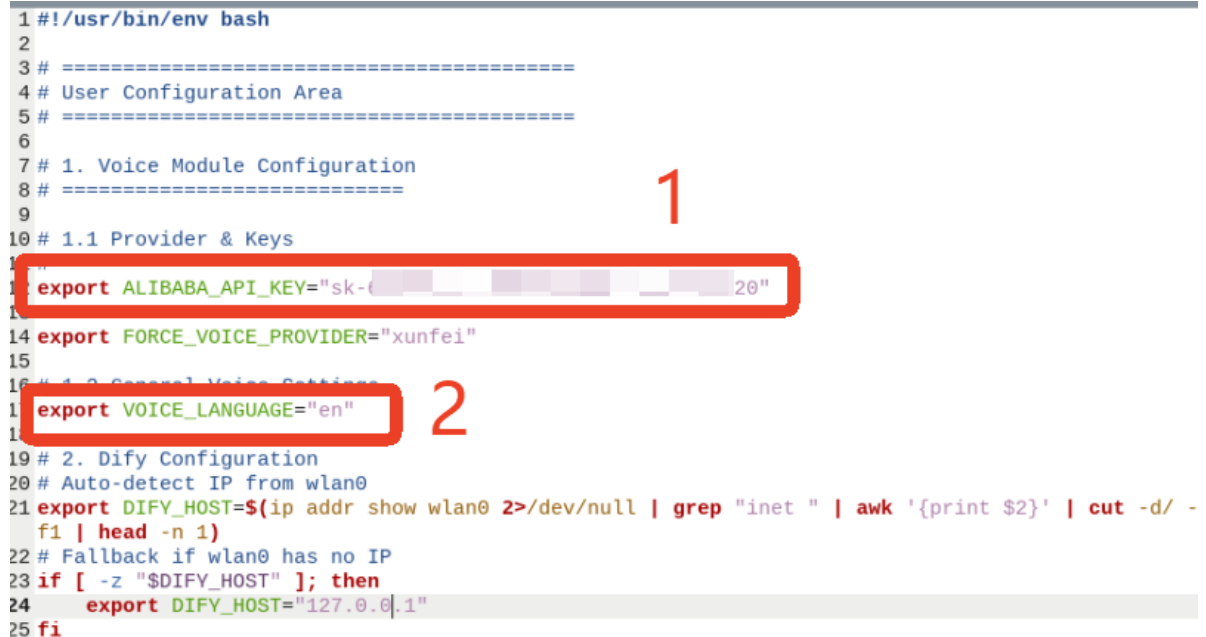
In this tutorial, we will learn how to issue commands via chat with Muto in the Dify interface.

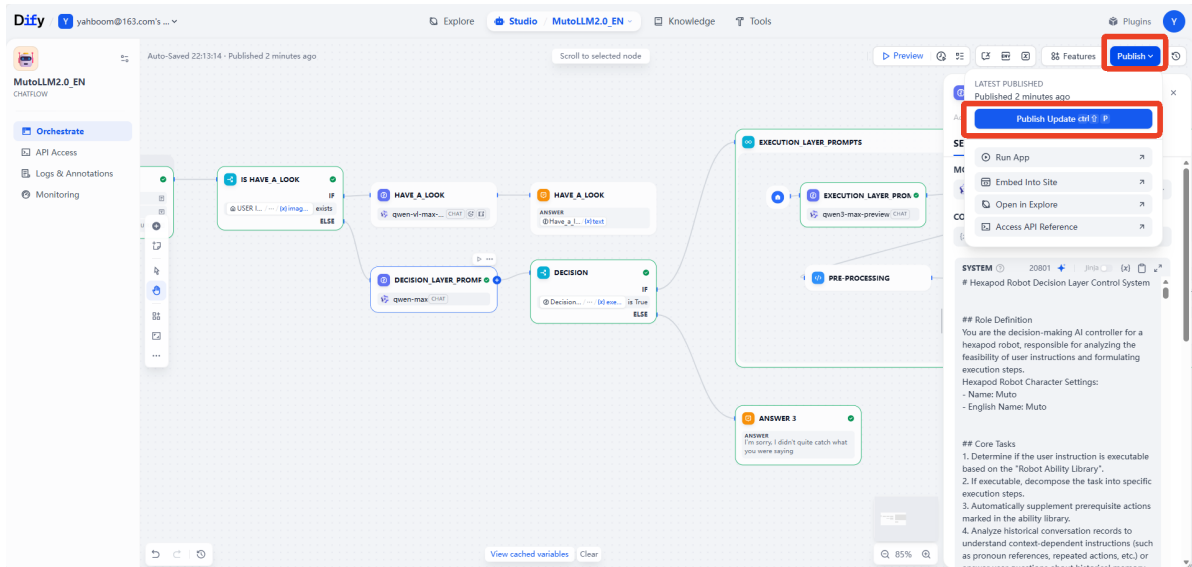
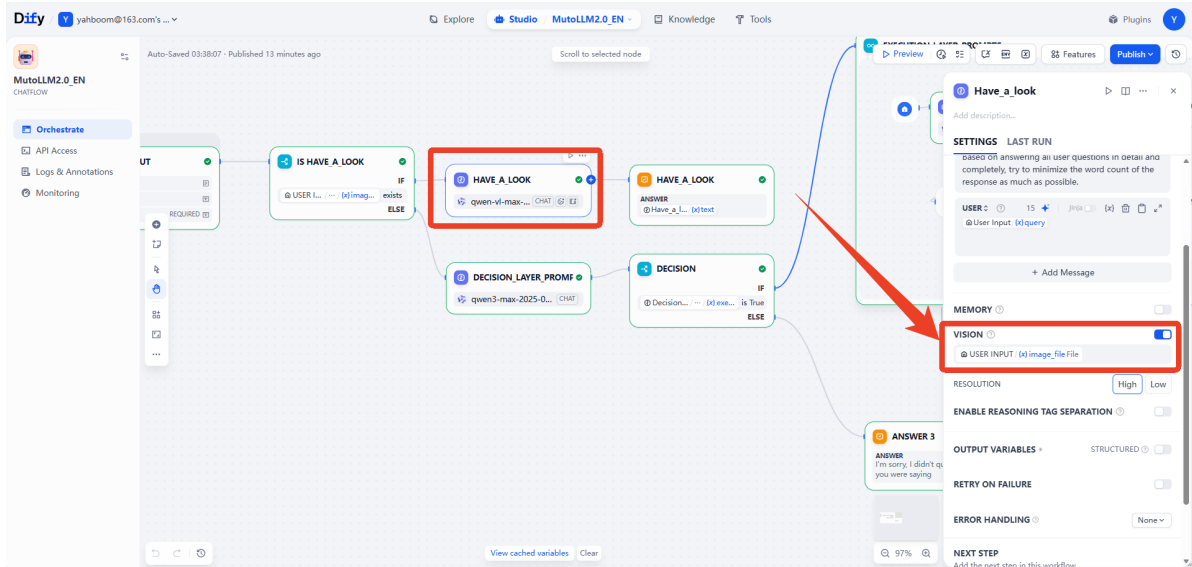
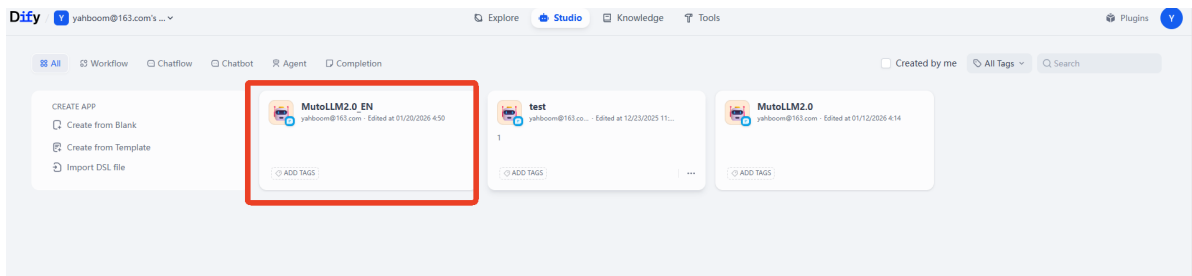
Operation Steps

Preparatory Steps

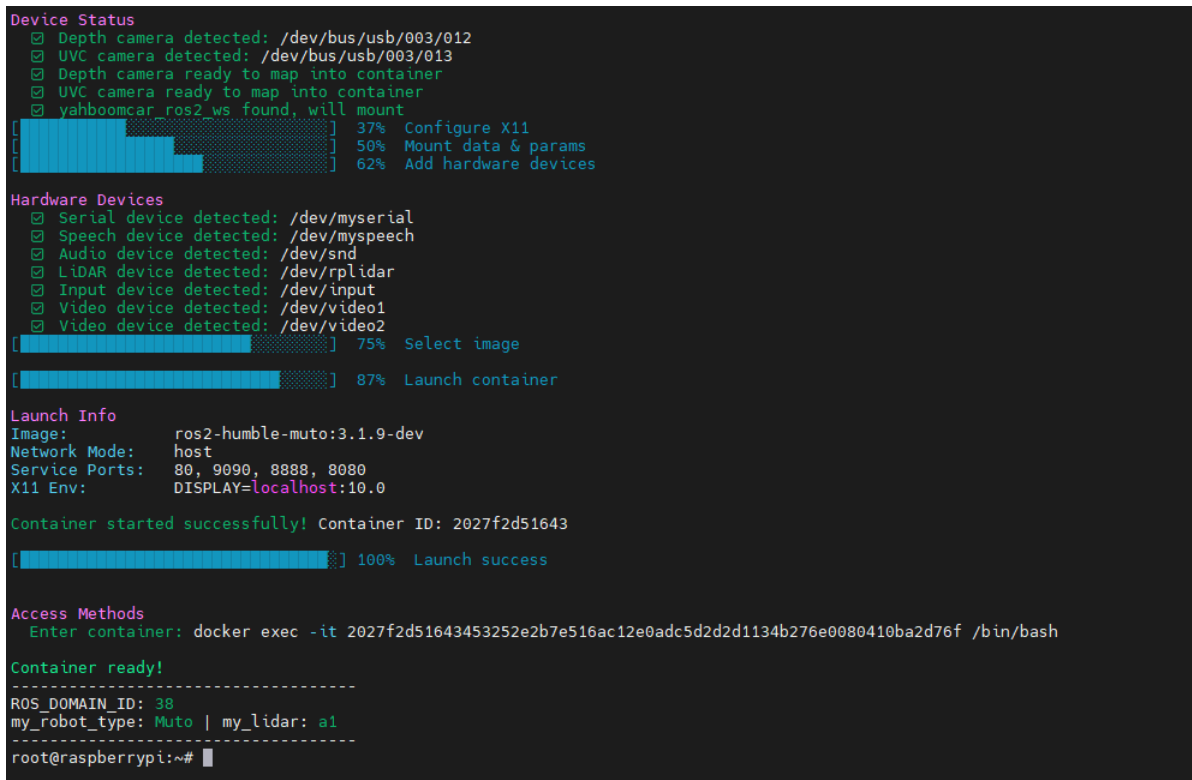
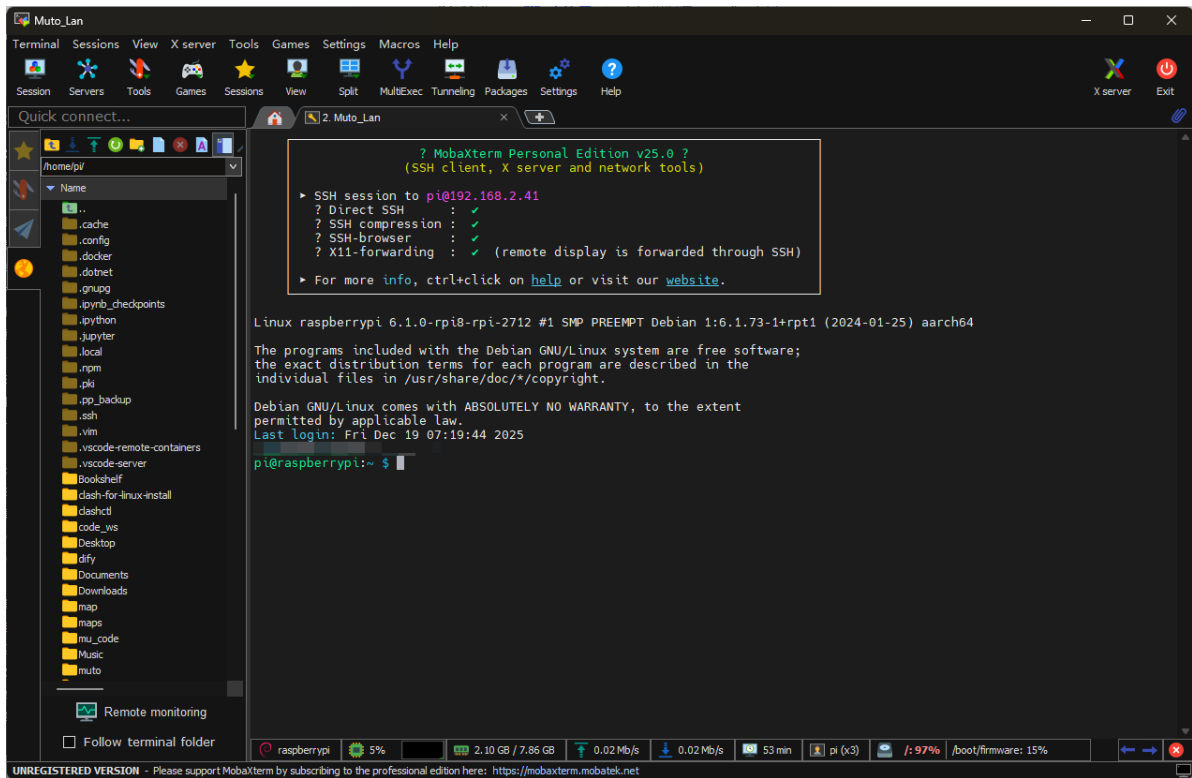
This lesson requires that the Dify terminal be launched beforehand, and all configuration items have been configured as in the previous section.

```
1 #!/usr/bin/env bash
2
3 # =====
4 # User Configuration Area
5 # =====
6
7 # 1. Voice Module Configuration
8 # =====
9
10 # 1.1 Provider & Keys
11 export ALIBABA_API_KEY="sk-20"
12
13 export FORCE_VOICE_PROVIDER="xunfei"
14
15 # 1.2 General Voice Settings
16 export VOICE_LANGUAGE="en"
17
18 # 2. Dify Configuration
19 # Auto-detect IP from wlan0
20 export DIFY_HOST=$(ip addr show wlan0 2>/dev/null | grep "inet " | awk '{print $2}' | cut -d/ -f1 | head -n 1)
21 # Fallback if wlan0 has no IP
22 if [ -z "$DIFY_HOST" ]; then
23     export DIFY_HOST="127.0.0.1"
24 fi
```





This lesson requires MutoRS to be powered on. Following the instructions in the previous chapters, start the large model terminal in Docker.



In this tutorial, we will use a plain text version of the large model, starting it with the `--no-voice` parameter. Using the `--no-voice` parameter eliminates the worry of being interrupted by external voice prompts during use!

```
cd muto-11m-2.0/ && ./rebuild_and_launch.sh --no-voice
```

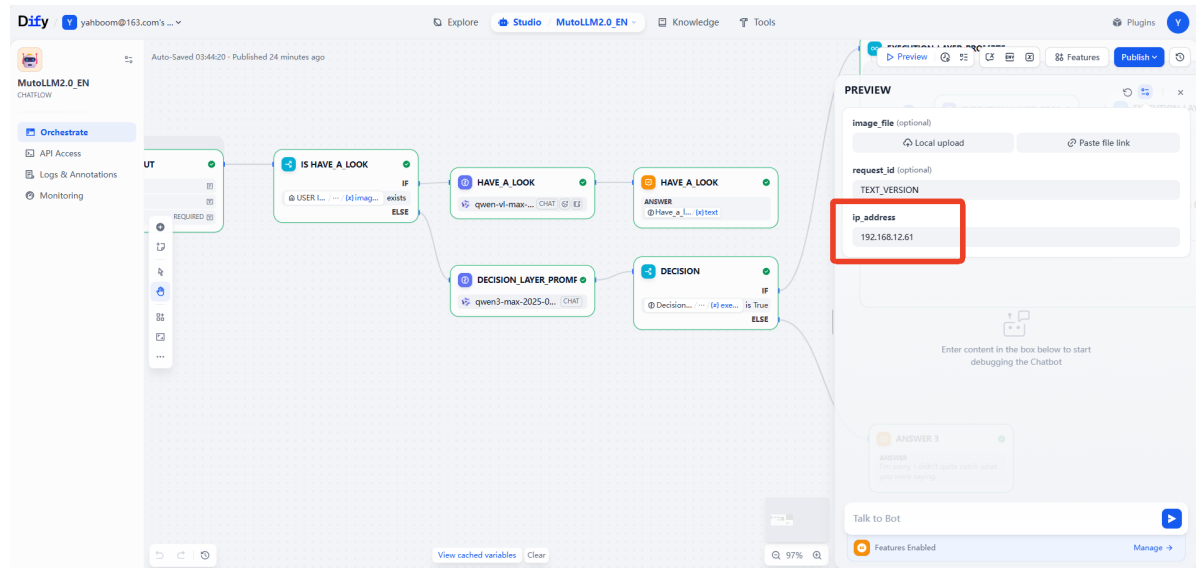
```

[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.voice_interface - INFO - Speech to text initialized successfully with alibaba provider
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.core.text_to_speech - INFO - TTS Provider alibaba
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - Audio queue manager initialized successfully
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - All voice modules initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315363643] [muto_controller_node]: Command executor initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315920470] [muto_controller_node]: Attempting to prestart camera system...
[muto_controller_node-1] X Node not running: /camera/camera
[muto_controller_node-1] Current running nodes: ['/muto_controller_node', '/navigation_manager']
[muto_controller_node-1] [INFO] [1766137697.078459054] [muto_controller_node]: Camera service started successfully: Camera started successfully
[muto_controller_node-1] [INFO] [1766137697.085229984] [persistent_image_capture_3188dc8d]: Persistent image capture node initialized for topic: /camera/color/image_raw with Best Effort QoS
[muto_controller_node-1] [INFO] [1766137697.086642747] [muto_controller_node]: Persistent camera node initialized: Camera node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.087635385] [muto_controller_node]: FastAPI application created successfully
[muto_controller_node-1] [INFO] [1766137697.149655757] [muto_controller_node]: API routes registered successfully
[muto_controller_node-1] 2025-12-19 09:48:17,149 - voice_module.core.voice_wake - INFO - Starting voice wake detector on port: /dev/myspeech
[muto_controller_node-1] 2025-12-19 09:48:17,154 - utils.serial_comm - INFO - Serial data receiving started
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.core.voice_wake - INFO - Voice wake detector started successfully
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.voice_interface - INFO - Wake detection started
[muto_controller_node-1] [INFO] [1766137697.155236161] [muto_controller_node]: Voice assistant auto-start: True
[muto_controller_node-1] [INFO] [1766137697.156658443] [muto_controller_node]: ROS2 publisher created successfully
[muto_controller_node-1] [INFO] [1766137697.157379009] [muto_controller_node]: Status timer and health check created successfully
[muto_controller_node-1] [INFO] [1766137697.157921559] [muto_controller_node]: Muto Controller Node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.158425850] [muto_controller_node]: Starting FastAPI server on 0.0.0.0:8080
[muto_controller_node-1] [INFO] [1766137697.161239218] [muto_controller_node]: FastAPI server started successfully
[muto_controller_node-1] INFO: Started server process [221]
[muto_controller_node-1] INFO: Waiting for application startup.
[muto_controller_node-1] INFO: Application startup complete.
[muto_controller_node-1] INFO: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)

```

Getting Started

For the large text model, you need to enter the Muto IP address here, which is the address used to access the Dify page.



For the navigation task, we also need to open multiple terminals.

First terminal, start the radar node

```
ros2 launch yahboomcar_nav laser_bringup_launch.py
```

```
----- rplidar_type = a1 -----
[INFO] [joint_state_publisher-1]: process started with pid [2811]
[INFO] [robot_state_publisher-2]: process started with pid [2813]
[INFO] [muto_driver-3]: process started with pid [2815]
[INFO] [sllidar_node-4]: process started with pid [2817]
[robot_state_publisher-2] [INFO] [1766387716.944917310] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-2] [INFO] [1766387716.953648152] [robot_state_publisher]: got segment base_link
[robot_state_publisher-2] [INFO] [1766387716.953696707] [robot_state_publisher]: got segment camera_link
[robot_state_publisher-2] [INFO] [1766387716.953711319] [robot_state_publisher]: got segment imu_link
[robot_state_publisher-2] [INFO] [1766387716.953720615] [robot_state_publisher]: got segment laser
[robot_state_publisher-2] [INFO] [1766387716.953729763] [robot_state_publisher]: got segment yh1_link
[robot_state_publisher-2] [INFO] [1766387716.953738707] [robot_state_publisher]: got segment yh2_link
[robot_state_publisher-2] [INFO] [1766387716.953748152] [robot_state_publisher]: got segment yh3_link
[robot_state_publisher-2] [INFO] [1766387716.953758096] [robot_state_publisher]: got segment yq1_link
[robot_state_publisher-2] [INFO] [1766387716.953767022] [robot_state_publisher]: got segment yq2_link
[robot_state_publisher-2] [INFO] [1766387716.953776059] [robot_state_publisher]: got segment yq3_link
[robot_state_publisher-2] [INFO] [1766387716.953785004] [robot_state_publisher]: got segment yz1_link
[robot_state_publisher-2] [INFO] [1766387716.953794503] [robot_state_publisher]: got segment yz2_link
[robot_state_publisher-2] [INFO] [1766387716.95380393] [robot_state_publisher]: got segment yz3_link
[robot_state_publisher-2] [INFO] [1766387716.953812133] [robot_state_publisher]: got segment zh1_link
[robot_state_publisher-2] [INFO] [1766387716.953821503] [robot_state_publisher]: got segment zh2_link
[robot_state_publisher-2] [INFO] [1766387716.953832726] [robot_state_publisher]: got segment zh3_link
[robot_state_publisher-2] [INFO] [1766387716.953841374] [robot_state_publisher]: got segment zq1_link
[robot_state_publisher-2] [INFO] [1766387716.953850985] [robot_state_publisher]: got segment zq2_link
[robot_state_publisher-2] [INFO] [1766387716.953859485] [robot_state_publisher]: got segment zq3_link
[robot_state_publisher-2] [INFO] [1766387716.953868096] [robot_state_publisher]: got segment z21_link
[robot_state_publisher-2] [INFO] [1766387716.953876670] [robot_state_publisher]: got segment z22_link
[robot_state_publisher-2] [INFO] [1766387716.953885670] [robot_state_publisher]: got segment z23_link
[sllidar_node-4] [INFO] [1766387716.991779363] [sllidar_node]: SLIDAR running on ROS2 package SLLidar.ROS2 SDK Version:1.0.
1, SLLIDAR SDK Version:2.0.0
[sllidar_node-4] [INFO] [1766387717.057315856] [sllidar_node]: SLLidar S/N: 6A97EDF9C7E29BD1A7E39EF2FA44431B
[sllidar_node-4] [INFO] [1766387717.057527393] [sllidar_node]: Firmware Ver: 1.29
[sllidar_node-4] [INFO] [1766387717.057576448] [sllidar_node]: Hardware Rev: 7
[sllidar_node-4] [INFO] [1766387717.108938542] [sllidar_node]: SLLidar health status : 0
[sllidar_node-4] [INFO] [1766387717.109190411] [sllidar_node]: SLLidar health status : OK
[sllidar_node-4] [INFO] [1766387717.345914003] [sllidar_node]: current scan mode: Sensitivity, sample rate: 8 Khz, max_distance: 12.0 m, scan frequency:10.0 Hz,
[joint_state_publisher-1] [INFO] [1766387718.130396704] [joint_state_publisher]: Waiting for robot_description to be published on the robot_description topic...
```

In the second terminal, we need to convert the radar data into odometry data for use by the navigation node.

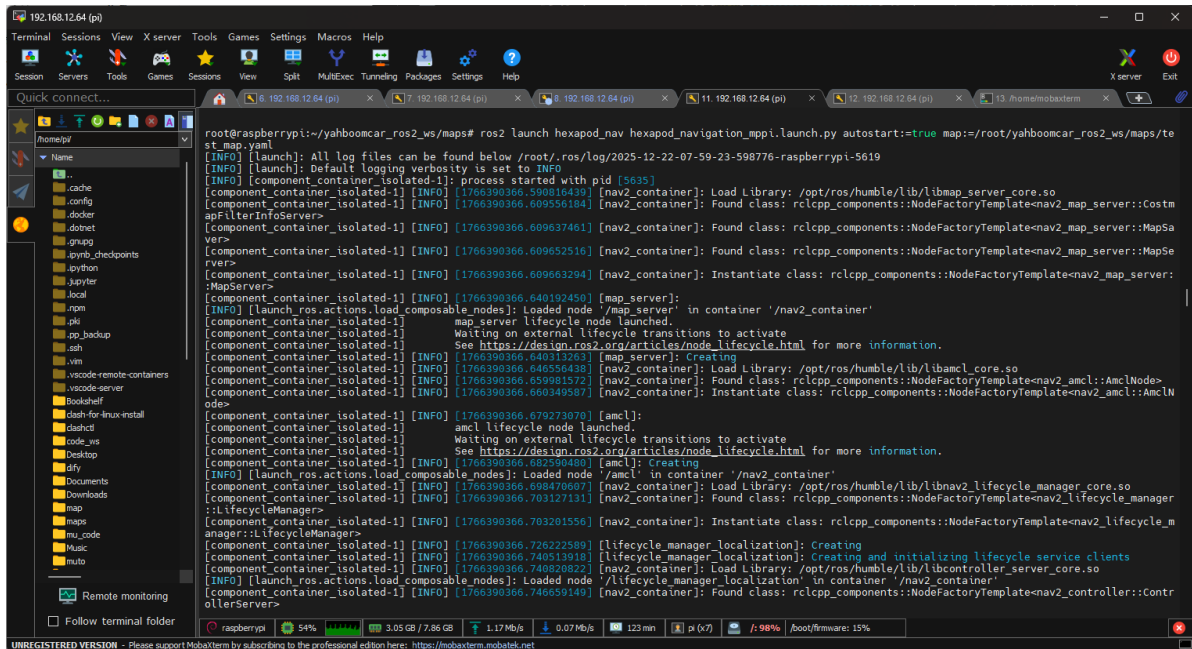
```
ros2 launch rf2o_laser_odometry rf2o_laser_odometry.launch.py
```

```
ROS_DOMAIN_ID: 38
my_robot_type: Muto | my_lidar: a1
root@raspberrypi:~/# ros2 launch rf2o_laser_odometry rf2o_laser_odometry.launch.py
[INFO] [launch]: All log files can be found below /root/.ros/log/2025-12-22-07-17-16-672206-raspberrypi-3030
[INFO] [rf2o_laser_odometry_node-1]: process started with pid [3034]
[rf2o_laser_odometry_node-1] [WARN] [1766387837.482827080] [rcl_logging_rosout]: Publisher already registered for provided node name. If this is due to multiple nodes with the same name then all logs for that logger name will go out over the existing publisher. As soon as any node with that name is destructed it will unregister the publisher, preventing any further logs for that name from being published on the rosout topic.
[rf2o_laser_odometry_node-1] [INFO] [1766387837.518134746] [rf2o_laser_odometry]: Initializing RF2O node...
[rf2o_laser_odometry_node-1] [WARN] [1766387837.690841774] [rf2o_laser_odometry]: Waiting for laser_scans....
[rf2o_laser_odometry_node-1] [WARN] [1766387837.650997738] [rf2o_laser_odometry]: Waiting for laser_scans....
[rf2o_laser_odometry_node-1] [INFO] [1766387837.657282326] [rf2o_laser_odometry]: [rf2o] Got first Laser Scan .... Configuring node
[rf2o_laser_odometry_node-1] [WARN] [1766387837.780888924] [rf2o_laser_odometry]: Waiting for laser_scans....
[rf2o_laser_odometry_node-1] [WARN] [1766387837.751088924] [rf2o_laser_odometry]: Waiting for laser_scans....
[rf2o_laser_odometry_node-1] [INFO] [1766387837.809141617] [rf2o_laser_odometry]: [rf2o] execution time (ms): 8.219006
[rf2o_laser_odometry_node-1] [INFO] [1766387837.809745856] [rf2o_laser_odometry]: [rf2o] LASERodom = [-0.031644 0.000000 3.140000]
[rf2o_laser_odometry_node-1] [INFO] [1766387837.809828374] [rf2o_laser_odometry]: BASEodom = [0.000000 -0.000000 0.000000]
[rf2o_laser_odometry_node-1] [WARN] [1766387837.83242103] [rf2o_laser_odometry]: Waiting for laser_scans....
[rf2o_laser_odometry_node-1] [WARN] [1766387837.900808983] [rf2o_laser_odometry]: Waiting for laser_scans....
[rf2o_laser_odometry_node-1] [INFO] [1766387837.981409297] [rf2o_laser_odometry]: [rf2o] execution time (ms): 29.811464
[rf2o_laser_odometry_node-1] [INFO] [1766387837.982392682] [rf2o_laser_odometry]: [rf2o] LASERodom = [-0.031714 -0.000601 3.140598]
[rf2o_laser_odometry_node-1] [INFO] [1766387837.982696959] [rf2o_laser_odometry]: BASEodom = [-0.000070 -0.000582 0.000598]
[rf2o_laser_odometry_node-1] [WARN] [1766387838.00879170] [rf2o_laser_odometry]: Waiting for laser_scans....
[rf2o_laser_odometry_node-1] [WARN] [1766387838.050920689] [rf2o_laser_odometry]: Waiting for laser_scans....
[rf2o_laser_odometry_node-1] [INFO] [1766387838.136603391] [rf2o_laser_odometry]: [rf2o] execution time (ms): 35.669350
[rf2o_laser_odometry_node-1] [INFO] [1766387838.137918719] [rf2o_laser_odometry]: [rf2o] LASERodom = [-0.031697 -0.000491 3.139621]
[rf2o_laser_odometry_node-1] [INFO] [1766387838.138022552] [rf2o_laser_odometry]: BASEodom = [-0.000054 -0.000503 -0.000379]
[rf2o_laser_odometry_node-1] [WARN] [1766387838.151004523] [rf2o_laser_odometry]: Waiting for laser_scans....
[rf2o_laser_odometry_node-1] [INFO] [1766387838.254072253] [rf2o_laser_odometry]: [rf2o] execution time (ms): 52.935045
[rf2o_laser_odometry_node-1] [INFO] [1766387838.254546363] [rf2o_laser_odometry]: [rf2o] LASERodom = [-0.031711 -0.000446 3.139621]
[rf2o_laser_odometry_node-1] [INFO] [1766387838.254610622] [rf2o_laser_odometry]: BASEodom = [-0.000067 -0.000457 -0.000379]
[rf2o_laser_odometry_node-1] [WARN] [1766387838.254974676] [rf2o_laser_odometry]: Waiting for laser_scans....
[rf2o_laser_odometry_node-1] [WARN] [1766387838.301996858] [rf2o_laser_odometry]: Waiting for laser_scans....
```

Next, we start the navigation node. Here we choose to use the MPPI planner, which performs better in narrow terrain.


```
ros2 launch hexapod_nav hexapod_navigation_mppi.launch.py autostart:=true map:=/root/yahboomcar_ros2_ws/maps/test_map.yaml
```

Please note that a pre-built map file (.yaml and .pgm, here we are using the map /root/yahboomcar_ros2_ws/maps/test_map) is required.



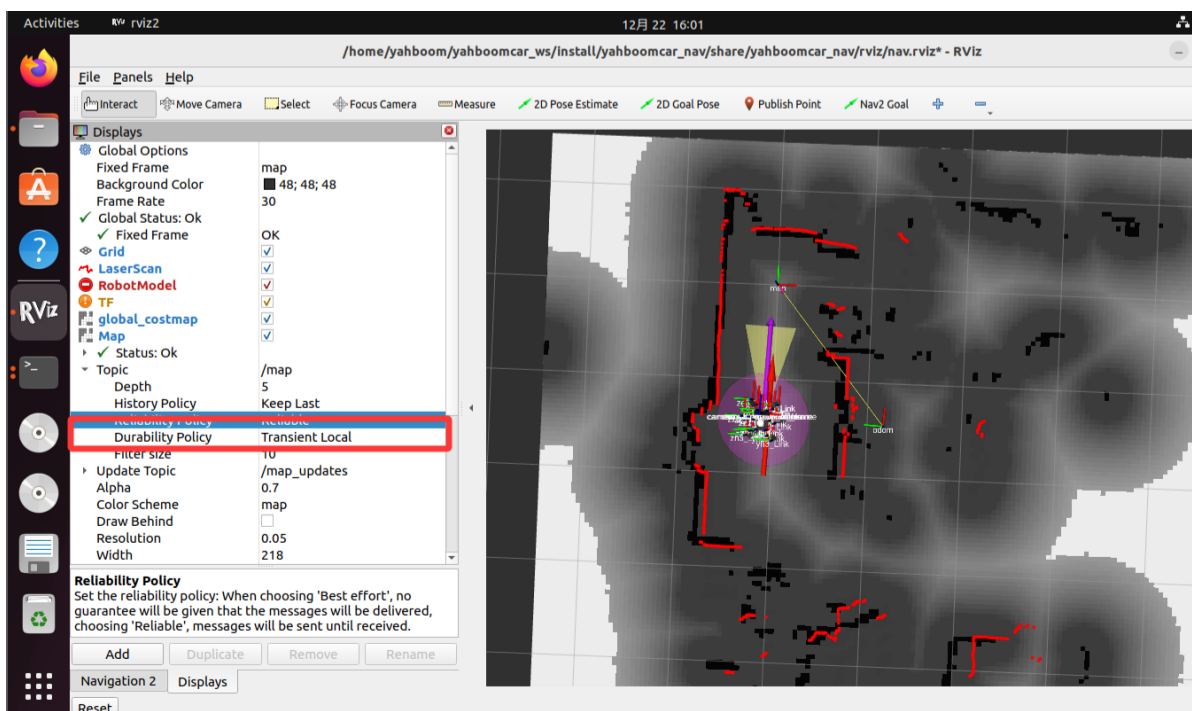
Finally, we can start the Rviz2 node to easily observe the navigation process.

This step is recommended to be performed in a VMware virtual machine!! **

```
ros2 launch yahboomcar_nav display_nav_launch.py
```

It is crucial to change the selected area in the image to **Transient Local**; otherwise, the map may not load.

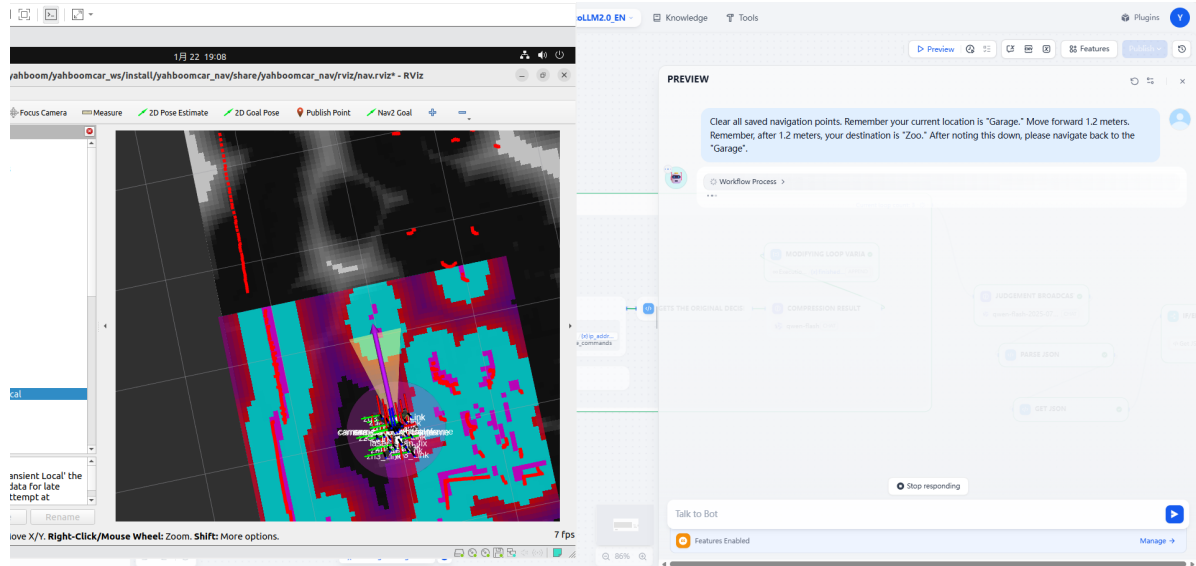
After selecting it, we use the 2D Pose Estimate tool to set the initial position.



After completing the above steps, we can use the large model to control Muto navigation!

We can enter the following commands:

Clear all saved navigation points. Remember your current location is "Garage."
Move forward 1.2 meters. Remember, after 1.2 meters, your destination is "Zoo."
After noting this down, please navigate back to the "Garage".



As you can see, navigation **may** fail. The large model will analyze the possible reasons for the failure based on the error message.

The error message for this failure is as follows:

```
[component_container_isolated-1] [INFO] [1766392121.987407276]
[controller_server]: Received a goal, begin computing control effort.
.....
[component_container_isolated-1] [INFO] [1766392121.994916498]
[controller_server]: Optimizer reset
[component_container_isolated-1] [WARN] [1766392123.016680997]
[BehaviorTreeEngine]: Behavior Tree tick rate 100.00 was exceeded!
[component_container_isolated-1] [INFO] [1766392123.087624461]
[controller_server]: Passing new path to controller.
.....
[component_container_isolated-1] [WARN] [1766392135.611962279]
[BehaviorTreeEngine]: Behavior Tree tick rate 100.00 was exceeded!
[component_container_isolated-1] [WARN] [1766392167.215183496]
[BehaviorTreeEngine]: Behavior Tree tick rate 100.00 was exceeded!
.....
[component_container_isolated-1] [INFO] [1766392167.220835206]
[global_costmap.global_costmap]: Message Filter dropping message: frame 'laser'
at time 1766392166.773 for reason 'the timestamp on the message is earlier than
all the data in the transform cache'
[component_container_isolated-1] [ERROR] [1766392167.349428846]
[controller_server]: Failed to make progress
[component_container_isolated-1] [WARN] [1766392167.349622410]
[controller_server]: [follow_path] [ActionServer] Aborting handle.
.....
[component_container_isolated-1] [WARN] [1766392167.375357691] [bt_navigator]:
[navigate_to_pose] [ActionServer] Aborting handle.
[component_container_isolated-1] [ERROR] [1766392167.375855844] [bt_navigator]:
Goal failed
```

This is because different geographical environments require different navigation configuration files.

We can modify the `hexapod_nav_mppi_params.yaml` configuration file to find suitable parameters for navigation based on the actual environment.

Manually Configuring Navigation Points

Besides letting Muto automatically record navigation points, we can also manually set them.

1. Configuration File Location

The navigation point configuration file **does not exist** by default. It will only be automatically generated by the system after you first call the save position function (e.g., `save_current_position`).

According to the code logic, the complete path of the generated file is:

```
~\muto-11m-2.0\packages\config\saved_positions.json
```

2. How to Manually Add Navigation Points

Since this file has not yet been generated, you can choose one of the following two methods:

Method 1: Let the System Generate Automatically (Recommended)

The simplest method is to have the robot perform a save operation at the current position (through voice commands or code calls). The system will automatically create the directory and file structure. Afterwards, you only need to open the file and modify the coordinates.

Method 2: Manually Create the File

1. Create a folder named `config` in the `~\muto-11m-2.0\packages\` directory.
2. Create a file named `saved_positions.json` inside the `config` folder.
3. Copy the following JSON template content into the file and modify it to your desired coordinate parameters.

File Content Template (`saved_positions.json`):

```
{
  "living_room": {
    "position": {
      "x": 1.5,
      "y": 2.0,
      "z": 0.0
    },
    "orientation": {
      "x": 0.0,
      "y": 0.0,
      "z": 0.0,
      "w": 1.0
    },
    "saved_time": 1705300000.0,
    "saved_time_str": "2024-01-15 12:00:00",
    "frame_id": "map",
    "child_frame_id": "base_link"
  }
}
```



```
},
"kitchen": {
  "position": {
    "x": -1.0,
    "y": 3.5,
    "z": 0.0
  },
  "orientation": {
    "x": 0.0,
    "y": 0.0,
    "z": 0.707,
    "w": 0.707
  },
  "saved_time": 1705300100.0,
  "saved_time_str": "2024-01-15 12:05:00",
  "frame_id": "map"
}
}
```

3. Parameter Description

- **Key (e.g., "living_room"):** The name of the navigation point, used in voice commands.
- **position:** The X/Y/Z coordinates of the target point (in meters).
- **orientation:** The orientation of the target point, represented by quaternions (x, y, z, w).
- **frame_id:** Usually fixed as "map", indicating a map-based coordinate system.